

Test before pre-selection (2024)

MCQ:

20x1=20

1. Which of the following is Autoboxing?

- a) `int a= new Integer(10);` b) `int a=10;`
c) `Integer ob=new Integer(10);` d) `int a =obj;`

2. Which of the following operation is invalid?

- a) Accessing static member from non static context b) accessing static member from static context
c) Accessing non static member from static context d) accessing non static element from static context

3. Which of the following access specifiers provides least accessibility?

- a) `public` b) `private`
c) `protected` d) `default`

4) Which of the following is an unary operator ?

- a) `>` b) `<`
c) `==` d) `!`

5) which is not a jump statement ?

- a) `return` b) `continue`
c) `System.exit(0)` d) `break`

6) Arrange in order of execution

- a) `++ ! / - >` b) `! / ++ - >`
c) `/ - ++ ! >` d) `/ + ! ++ -`

7. The condition `!((2>5 && 2<4) || 2!=3)` evaluates to

1. `true` 2. `false`
3. 😊 4. `error`

8. `while (!(2!=3)) System.out.print("hi");`

- a) `error` b) `hi`
c) `will not execute` d) `hi hi`

9. Accessing an element at non existing index in array results as

- a) Nothing
- b) ArrayIndexOutOfBoundsException
- c) ArrayIndexOutOfBoundsException
- d) Not found

10. Which of the statements is true

- a) all characters can be converted to byte
- b) all integers can be converted to characters
- b) not all characters can be converted to integers (int)
- d) not all integers can be converted to character

11. String s="AARUSH"; Size of s is-

- a) 15 bytes
- b) 10 bytes
- c) 14 bytes
- d) 12 bytes

12. Which one is correct

- a) byte a=20 *10;
- b) char ch= (char)(20*10);
- c) int k= 20**"2"
- d) boolean b= (a>1) ? true: false;

13) trim() removes spaces from a String literal

- a) true
- b) partially true
- b) false
- d) 😊

14) which is true

- a) a function can have multiple return statements and can return multiple values
- b) a function can have single return statement and can return multiple values
- c) a function can have multiple return statements but can return only one value
- d) a function can only have one return statement and can return only one value

15) Size of array is N then last index of array is

- a) N-1
- b) N+1
- c) N
- c) N.length

16) Integer.valueOf() returns

- a) Integer object
- b) data of type int
- b) String
- d) character

17. To find an element in an array of size N using linear search, the maximum number of loops possible is-

- a) N-1 b) N+1
- b) N d) N/2

18) An array containing five elements of type double will occupy

- a) 40 bytes b) 42 bytes
- c) 41 bytes c) 20 bytes

19. s.charAt(s.length())

- a) will execute b) won't execute
- b) execute but no result d) 😊

20. extension of byte code created by JAVA compiler is

- a) .java b) .class
- b) .file d) .binary

SAQ. Answer the following

10x2=20

1. Define Method Signature

2. Math.max("Hello".length(), "Bye".length()) + "Bye".charAt(0)

3. int k=5, j=9

```
k+= k++ - ++j +k;
System.out.println("j=" +j);
System.out.println("k=" +k);
```

4. double d= 5 +2/3 *6

```
System.out.println(d);
```

5. char ch1= 'a';

```
int n= ch1-32;
System.out.println(n);
System.out.println((char)n);
```

6. Write $(a+b)^n / \sqrt{2as + u^2}$ using JAVA

7. Write prototype of a function that takes a character and returns an integer

8. (2>3 || 3<4) && 5!=3 find the returning boolean value

9. for (int i=2; i<=10; i+=2){

```

    if( i==2) continue;
    if(i>7) break;
    System.out.println(i);
}
10. int k=3, i=2;
    while(i<=5){
        i++;
        if(i==2) continue;
        k*=i;
    }
    System.out.println(k);
    Find output and number of times the loop executes.

```

Section B

15x4=60

- Write a program to take an array from user in ascending order. int a number (s) search for it using binary search.
- Write a program to find discount and amount to be paid with the following specifications

class name : Shop

data members: int price, dis, ap;

Member methods: Shop (int price) default constructor to initialize price with parameterized price and dis=0 and ap=0

void discount() to calculate discount and amount to be paid

void PrintResult() to print discount and amount to be paid

Purchase Amount	Discount Percentage
Upto 5000	10%
5001 to 10000	15%
10001 to 15000	20%
Above 15000	25%

- Write a program to overload a function perimeter() to calculate and perimeter of circle, square and rectangle. call functions based on user's choice

- Write a program to take a String from user and convert each vowel to UpperCase

Sample Input: abracadabra

Sample output AbrAcAdAbA

- Write a menu driven program to print the following based on user's choice.

a) 0 1 1 2 3 5 8 upto n terms (n is given by user) (Fibonacci Series)

b) $S = \frac{a^2}{3} + \frac{a^3}{4} + \frac{a^4}{5} + \frac{a^5}{6} + \frac{a^6}{7} + \frac{a^7}{8} + \frac{a^8}{9} + \dots$ + upto nth term

- Take number from user and check

a) if it's palindrome number or not (reverse is same as original)

b) if it's perfect number or not (Perfect number: sum of factors other than itself is equal to original number.)